

Bayesian Decision Tree with Monte Carlo Simulation

Theory, Implementation, and YAML Configuration Guide

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Abstract

This document describes the theory, implementation, and usage of `bayes-tree-eng.py`, a tool that encodes hypotheses as Bayesian decision trees and uses Monte Carlo simulation to propagate uncertainty. Top-level branches act as independent evidence combined at the root via likelihood ratios in log-odds space. The tool produces posterior distributions, sensitivity analyses, and importance rankings.

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1 Introduction

The Bayesian Decision Tree tool combines Bayesian inference with Monte Carlo simulation to propagate uncertainty through a tree of evidence. Each branch of the tree represents an independent piece of evidence characterized by a *likelihood ratio* (LR), and the tool computes a posterior probability distribution for the root hypothesis.

The motivation is to provide a structured, transparent framework for combining multiple heuristic likelihood ratios into a posterior probability. It is useful for exploratory “what-if” analysis, sensitivity studies, and producing an interpretable tree of how local evidence shifts beliefs.

2 Theoretical Background

2.1 Bayes’ Theorem

For a hypothesis H and evidence E :

$$P(H|E) = \frac{P(E|H) P(H)}{P(E)} \quad (1)$$

The **likelihood ratio** (LR) form is:

$$\text{LR} = \frac{P(E|H)}{P(E|\neg H)} \quad (2)$$

2.2 Log-Odds Representation

Converting probability to log-odds:

$$\text{lo}(p) = \ln \frac{p}{1-p} \quad (3)$$

And back:

$$p = \frac{1}{1 + e^{-\text{lo}}} \quad (4)$$

A Bayesian update in log-odds space is simply additive:

$$\text{lo}(P(H|E)) = \text{lo}(P(H)) + \ln(\text{LR}) \quad (5)$$

This additive property makes it natural to combine multiple independent pieces of evidence.

2.3 Combining Independent Evidence

Given n independent evidence branches with likelihood ratios $\text{LR}_1, \text{LR}_2, \dots, \text{LR}_n$, the combined posterior in log-odds is:

$$\text{lo}(\text{posterior}) = \text{lo}(\text{prior}) + \sum_{i=1}^n \ln(\text{LR}_i) \quad (6)$$

Equivalently, in odds form:

$$\text{odds}(H \mid E_{1:n}) = \text{odds}(H) \prod_{i=1}^n \text{LR}_i \quad (7)$$

The tool computes this at the root level, treating each top-level child as an independent evidence source.

2.4 Effective Likelihood Ratio

The *effective LR* is the single likelihood ratio that would produce the same posterior from the given prior:

$$\text{LR}_{\text{eff}} = \exp(\log(\text{posterior}) - \log(\text{prior})) \quad (8)$$

2.5 Monte Carlo Uncertainty Propagation

When the LR for each branch is uncertain (specified as an interval rather than a point value), the tool uses Monte Carlo simulation:

1. For each simulation run, sample an LR from each branch's distribution.
2. Compute the combined posterior using the log-odds sum.
3. Repeat N times (default: 10,000) to build a posterior distribution.
4. Report summary statistics: median, mean, 90% credible interval.

2.6 LR Sampling Distributions

Three distributions are available for sampling LR values from the interval $[\text{lr_min}, \text{lr_max}]$:

log_uniform (default) — Uniform in log-space. Best when uncertainty spans orders of magnitude. The sampled value is:

$$\text{LR} = \exp(\text{Uniform}(\ln(\text{lr_min}), \ln(\text{lr_max}))) \quad (9)$$

uniform — Uniform in linear space:

$$\text{LR} = \text{Uniform}(\text{lr_min}, \text{lr_max}) \quad (10)$$

beta — Beta distribution scaled to the interval, controlled by parameters **lr_alpha** and **lr_beta**:

$$\text{LR} = \text{lr_min} + \text{Beta}(\alpha, \beta) \times (\text{lr_max} - \text{lr_min}) \quad (11)$$

3 Tree Structure and Chaining

The tree has two distinct roles:

- **Root level:** Children are combined as independent evidence (log-odds sum). Each child contributes its own LR to the root posterior.
- **Sub-levels:** Children represent drill-down or decomposition of the parent evidence. The child's prior is set to the parent's posterior (chaining), showing how the probability evolves through successive reasoning steps.

Importantly, children of top-level branches are *not* multiplied into the root combination. They only serve to explain or decompose a branch's LR. If child-level evidence should aggregate to the root, the model must be restructured so that the relevant nodes appear at the top level.

4 Sensitivity Analysis

The tool provides two sensitivity measures:

1. **Threshold probabilities:** $P(\text{posterior} < 5\%)$, $P(\text{posterior} < 10\%)$, $P(\text{posterior} > 50\%)$.
2. **Leave-one-out ranking:** For each branch, compute the posterior without that branch. The difference from the baseline median shows how much each branch influences the result.

5 YAML File Structure

The input is a YAML file defining a tree of evidence. Below is the complete specification.

5.1 Root Node

```
node "Hypothesis_name"
prior 0.5 # Prior probability (0 to 1)
children
  - ... # List of evidence branches
```

5.2 Evidence Node

Each child node represents one piece of evidence:

```
- node "Evidence_description"
  evidence_type for # "for", "against", or "neutral"

  # Option A: Point estimate
  likelihood_ratio 2.5 # Exact LR value

  # Option B: Uncertainty interval (preferred)
  lr_min 0.01 # Lower bound of LR
  lr_max 0.10 # Upper bound of LR
  lr_dist log_uniform # Distribution: log_uniform |
    uniform | beta

  # Optional for beta distribution
  lr_alpha 2 # Beta shape parameter alpha
  lr_beta 2 # Beta shape parameter beta

  # Optional sub-evidence (drill-down)
  children
    - node "Sub-evidence"
      evidence_type for
      likelihood_ratio 1.5
```

5.3 Evidence Type

for — Evidence supporting the hypothesis ($LR > 1$).

against — Evidence against the hypothesis ($LR < 1$).

neutral — Uninformative evidence ($LR \approx 1$).

The tool validates that `evidence_type` is consistent with the LR values and warns if there is a mismatch.

5.4 Likelihood Ratio Interpretation

LR value	Meaning
$LR \gg 1$	Strong evidence for the hypothesis
$LR > 1$	Moderate evidence for
$LR = 1$	Neutral (no update)
$LR < 1$	Evidence against
$LR \ll 1$	Strong evidence against

6 Complete YAML Example

```
node "Patient_has_disease_X"
prior 0.10
children
  - node "Positive_blood_test"
    evidence_type for
    lr_min 3.0
    lr_max 8.0
    lr_dist log_uniform
    children
      - node "Test_specificity_concerns"
        evidence_type against
        lr_min 0.5
        lr_max 0.9

  - node "Family_history_present"
    evidence_type for
    likelihood_ratio 2.0

  - node "Age_below_typical_onset"
    evidence_type against
    lr_min 0.3
    lr_max 0.7
    lr_dist uniform

  - node "Symptom_pattern_matches"
    evidence_type for
    lr_min 1.5
    lr_max 4.0
    lr_dist beta
    lr_alpha 2
    lr_beta 3
```

7 Usage

7.1 Running the Tool

```
# Default: 10,000 simulations
python bayes-tree-eng.py examples/mycase.yaml

# Custom number of simulations
```

```
python bayes-tree-eng.py examples/mycase.yaml 50000
```

7.2 Output Sections

The tool produces the following output:

1. **Validation warnings** — Flags any inconsistencies between `evidence_type` and LR values.
2. **Combined posterior** — Summary statistics (median, mean, CI) and ASCII histogram of the posterior distribution.
3. **Tree view** — Hierarchical display showing each branch’s contribution, with prior $\rightarrow$ posterior transitions.
4. **Sensitivity analysis** — Threshold probabilities.
5. **Importance ranking** — Leave-one-out analysis showing which branches have the most influence.

7.3 Tips for Creating YAML Files

1. **Start with the prior.** What is the base rate or initial probability before any evidence is considered?
2. **List independent evidence.** Each top-level child should be logically independent of the others.
3. **Prefer intervals over point estimates.** Using `lr_min` and `lr_max` captures your uncertainty about the strength of evidence.
4. **Use `log_uniform` for uncertain LRs.** When you know the LR is “somewhere between 0.01 and 0.1” but aren’t sure where, log-uniform gives appropriate weight to different orders of magnitude.
5. **Add children for drill-down.** Sub-nodes explain or decompose a parent’s evidence but don’t add to the root combination.
6. **Validate your intuitions.** Run the tool and check that the output matches your qualitative expectation. If not, reconsider your LR estimates.

8 Limitations and Assumptions

- **Independence assumption.** Top-level branches are assumed conditionally independent given H and $\neg H$. This is a strong assumption and should be scrutinized when modeling correlated evidence. In practice, if two branches share a common information source, they should be merged or one should be made a sub-node of the other.
- **Children do not aggregate to root.** Children are used only for chained drill-down display. If child-level evidence should contribute to the root posterior, the model must be restructured so that the relevant nodes appear at the top level.
- **Log-uniform bias.** The default log-uniform sampling biases toward multiplicative scales. Users should choose a distribution that reflects their actual beliefs about where in the interval the true LR lies.

- **Subjective LR estimates.** The tool is only as good as the likelihood ratios you assign. It makes subjective reasoning transparent and quantitative, but does not eliminate subjectivity.
- **Numerical stability.** The implementation clamps probabilities and log-odds to safe numerical ranges to avoid overflow, which may slightly affect extreme posteriors (very close to 0 or 1).

9 Implementation Notes

- **Log-odds stability:** Probabilities are clamped to $[10^{-12}, 1 - 10^{-12}]$ before log-odds conversion; log-odds are clamped to $[-700, 700]$ before exponentiation.
- **Sampling:** The same random draw strategy is applied per simulation; users can set simulation counts via the command line.
- **Tree view:** A recursive collection routine computes node-level posteriors by feeding a parent's posterior median as prior into children, producing local medians and credible intervals.
- **Effective LR for root:** The root node's LR interval is derived from the 5th–95th percentiles of the effective LR distribution across all Monte Carlo runs.

10 Dependencies

The Python script requires:

- Python 3.8+
- PyYAML (`pip install pyyaml`)